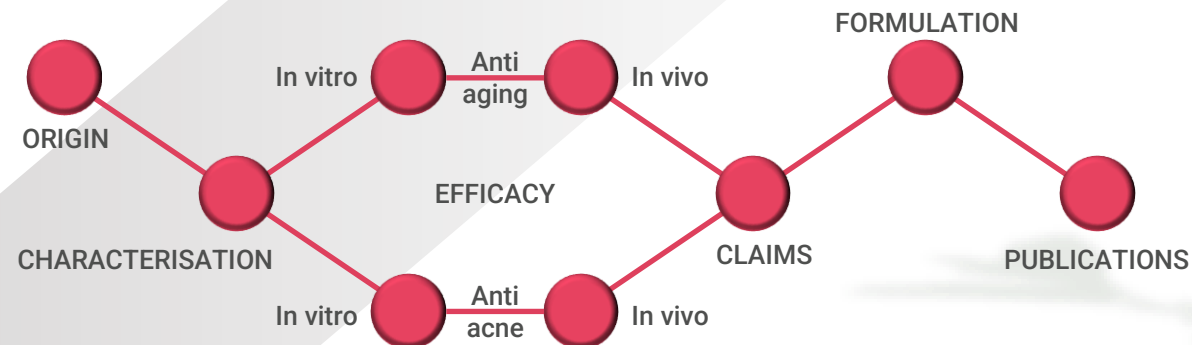


Sytenol[®] A

The true alternative to retinol



Sytenol® A: The Digital Buzz

Since 2018,
major buzz
around Bakuchiol

- Mintel listed Bakuchiol among the top trends in 2019 and beyond

MINTEL

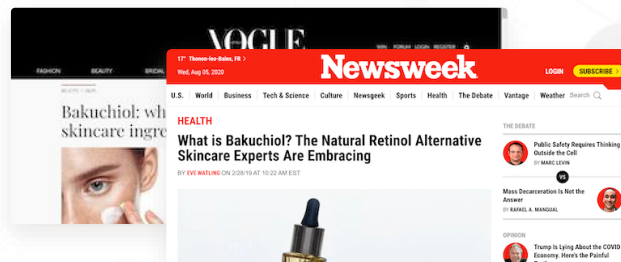
➤ SEE SOURCE

On Google Trends, the popularity of « bakuchiol » increased dramatically



Multiplication of articles:
Sytenol® A is the most
wanted active ingredient
in the world

KEY OPINION LEADERS



YOUTUBE VIDEOS



BAKUCHIOL VS RETINOL? ACNE & ANTI AGING| DR DRAY
Dr Dray • 60 k vues • il y a 11 mois
skincare #acne #antiaging BAKUCHIOL VS RETINOL? DR DRAY REVIEWS BAKUCHIOL FOR ACNE & ANTI AGING.



RETINOIDS | Retinoic Acid, Retinal, Retinol, HPR + Bakuchiol
Gothamista • 259 k vues • il y a 1 an
This video is all about Vitamin A! I'm exploring the different types of Retinoids and sharing the products I've used and loved from ...



I Tried the HERBIVORE Bakuchiol Retinol Alternative SERUM For 30 Days | Here's My Honest Review
Britt Wilkin • 15 k vues • il y a 10 mois
Is the Herbivore Bakuchiol Retinol Alternative serum worth it? I finally have this one ready for you and ...



The Game Changer: British J Dermatology Publication

The logo for the British Journal of Dermatology (BJD), consisting of the letters 'BJD' in white serif font on a red square background.

Bakuchiol (Sytenol® A) was shown in a recent study published in British J Dermatology to match the established benefits of retinol, with a better skin tolerance

Further to this publication, there has been a strong emphasis on Bakuchiol from key opinion leaders

 [SEE PUBLICATION](#)

PLAIN LANGUAGE SUMMARY

British Journal of

Assessment of topical bakuchiol and retinol for facial photoageing

S. Dhaliwal, I. Rybak, S.R. Ellis, M. Notay, M. Trivedi, W. Burney, A.R. Vaughn, M. Nguyen, P. Rees, S. Bosanac, H. Yan, N. Foolad and R.K. Sivamani

This summary relates to <https://doi.org/10.1111/bjd.16918>

British Journal of Dermatology, **180**, 289–296, February 2019

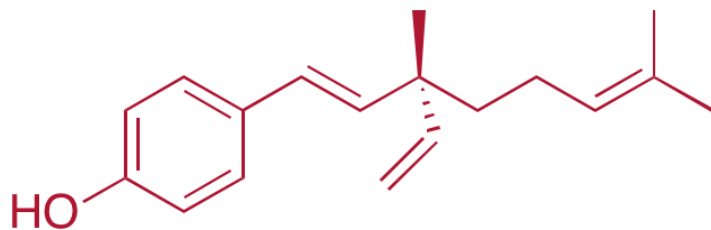
Summary

Bakuchiol is found mainly in the seeds of the Indian plant *Psoralea corylifolia* (babchi) and has recently been shown to have a number of antioxidant and anti-inflammatory properties. The goal of this study, by researchers in California, Michigan, Florida and Pennsylvania in the US, was to compare the efficacy and safety of bakuchiol with the commonly-used anti-ageing ingredient retinol. 44 volunteers were asked to apply bakuchiol 0.5% cream twice daily, or retinol 0.5% cream daily, to facial skin for 12 weeks. Retinol 0.5% has been shown to be effective at preventing and addressing signs of skin ageing but can have side effects such as stinging, scaling and redness. A facial photograph and analytical system was used to take and analyse photographs of patients at 0, 4, 8 and 12 weeks of the study. Patients also answered questions about



Sytenol® A: The Benefits of Retinol without the Drawbacks

Sytenol® A

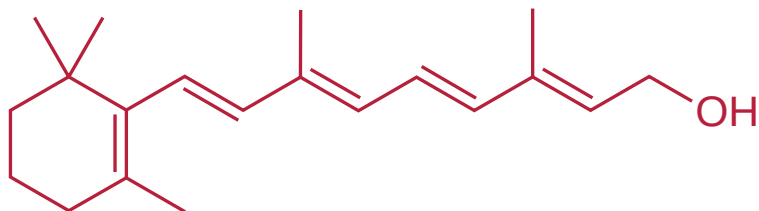


Sytenol® A is
equally effective
in addressing
aging skin and is:

- ✓ Devoid of safety issues
- ✓ Easy to formulate
- ✓ Devoid of hydrolytic stability problem
- ✓ Photostable (*day application possible*)

Retinol

First generation



- > Reversing aging
- > Protecting skin from further damage
- > Improving acne skin



- > **Side effects:** Skin irritation, skin sensitization and teratogenicity
- > **Photochemical and hydrolytic instability:** use primarily restricted to night-time only
- > **Incompatibility:** difficulty in formulating products



Sytenol® A: Product Overview

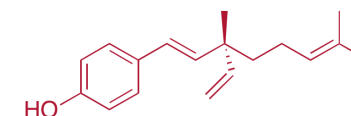
- > **INCI name: Bakuchiol**
- > **100% Natural**
extracted and purified from
Psoralea corylifolia (babchi seeds)
- ⚠ > **99% purity of bakuchiol ≠ Babchi oil**
- > **Appearance:** Yellowish or
Reddish yellow viscous liquid
- > **Oil Soluble**
- > **Use level:** 0.5 to 1.0%



SEE DETAIL



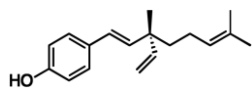
Jan, 1st 2021



Sytenol® A: Product Overview

Bakuchiol

- > Bakuchiol (INCI) is minimum 99% (Sytenol® A)
 - CAS # 10309-37-2
 - REACH # 01-2120113117-71-0000
- > Psoralens content < 25ppm (typical value < 10 ppm)
- > Residual solvents < 25ppm (typical value < 10 ppm)



- > Antiaging
- > Acne-prone skin

To learn more: www.bakuchiol.net

SAFE FOR TOPICAL USE

Babchi oil / Bakuchi oil

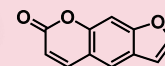


- > Babchi or Psoralea Corylifolia oil are undefined materials that contain minor amount of Bakuchiol (from 1 to 12%)
- > Activity based on Skin photosensitizers (psoralene/isopsoralene)
- > Possibly high hexane residual content in refined oils. For babchi oils, INCI name must be psoralea corylifolia plant or seed extracts

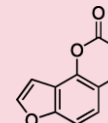
≠



Babchi oil



Psoralene



Isopsoralene

- > Vitiligo treatment
Psoriasis
(PUVA-therapy)

PHOTOTOXIC AND NOT STANDARDIZED: Unsuitable for cosmetic use

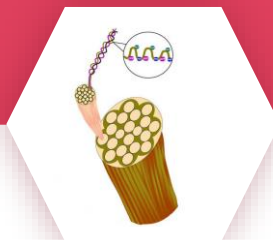
Sytenol® A: Mechanisms for Restorative & Preventative Antiaging

RETINOL-LIKE ACTIVITY



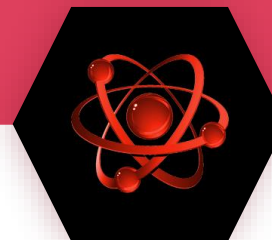
- + Volcano plots
- + Retinoid Binding & Metabolizing Genes
- + Extracellular Matrix Genes
- + Dermo-epidermal Junction Genes

COLLAGEN STIMULATOR



- + Stimulation of Collagen I, III & IV

DEFENSE AGAINST OXIDATIVE STRESS



- + Broad spectrum scavenger
- + Stabilizer for Retinol

DEFENSE AGAINST INFLAMMATORY STRESS



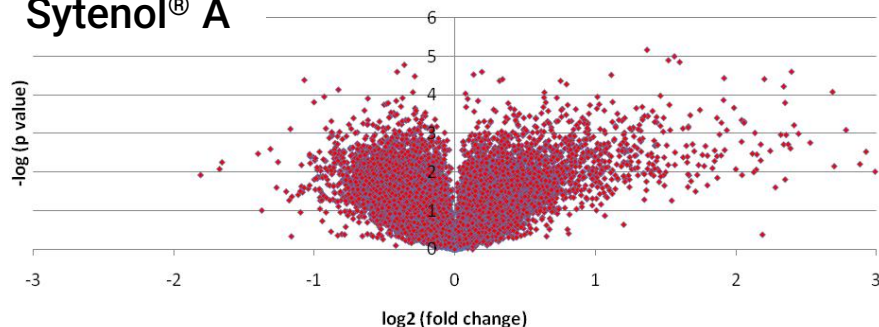
- + Inhibition of multiple genes and markers of inflammation

... Protective antiaging ...

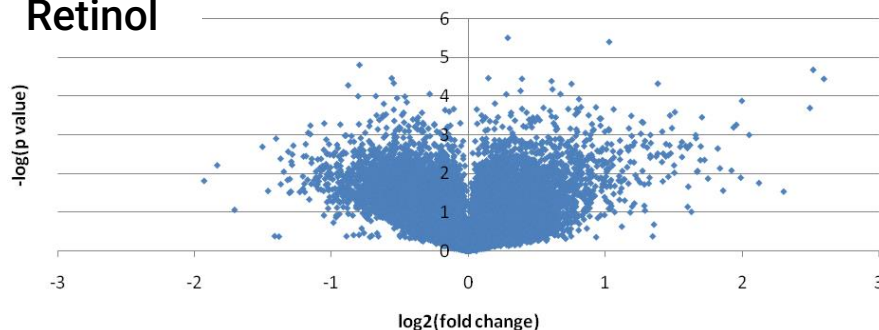
... Restorative antiaging ...

Sytenol® A vs Retinol: Volcano Plots

Sytenol® A



Retinol



DNA microarray study done using full thickness Epiderm tissue from Mattek

Scatter points results in two regions of interest in the volcano plot: those points that are found towards the top of the plot and that are far to either the left- or the right-hand side. These represent values that display large magnitude fold changes (hence being left- or right- of center) as well as high statistical significance (hence being towards the top).



Comparative Volcano plots of Sytenol® A and Retinol show dramatic similarities in gene expression profiles of the two compounds indicating similar functional analogy

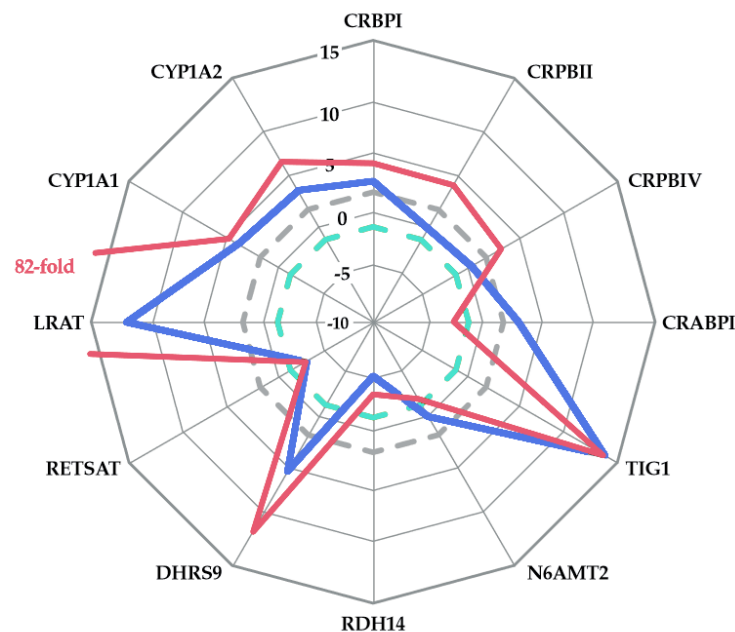
Sytenol® A: Modulator of Retinoid Binding & Metabolizing genes

Gene expression profile: Fold-change vs Control

SEE PROTOCOL & GENE DETAILS

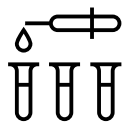
- Retinol
- Threshold of stimulatory activity
- Sytenol® A
- Threshold of stimulatory activity

$p < 0.05$



Modulation of retinoid binding & metabolizing genes is essential for Sytenol® A to function like Retinol

Sytenol® A: Modulator of Retinoid Binding & Metabolizing genes



Protocol

DNA microarray: Used full thickness Epiderm tissue from Mattek;
Considered expression profile of over ± 2 -fold change with $P < 0.05$

- > **Cellular retinol binding protein (CRBP):**
Indispensable for Retinol & Retinoic acid biosynthesis and storage of Retinol.
- > **Cellular retinoic acid binding protein (CRABP):**
Downregulation by Sytenol® A eliminates Retinoic acid-induced toxicity.
- > **N-6 adenine-specific DNA methyltransferase 2 (N6AMT2):** Retinoic acid resistance might be combatted by the use of epigenetic modifying agents such as DNA methyl transferase inhibitors. Down-regulation of N6AMT2 by Sytenol® A may reduce Retinoic acid-induced toxicity.
- > **Tazarotene-inducible gene 1 (TIG1):** TIG1 is down-regulated in acne, rosacea and psoriasis. Up-regulation by Sytenol® A expects to provide skin benefits.
- > **Retinol dehydrogenase/reductase (RDH14; DHRS9; RETSAT):**
Converts Retinol to Retinal and then to retinoic acid. Retinol saturase (RETSAT) is involved in normalization of adipocyte differentiation. Lecithin-retinol acyltransferase (LRAT): Retinol esterification with long-chain fatty acid by LRAT is the key step in Retinol absorption & storage.
- > **Cytochrome P450 (CYP1A1; CYP1A2):** P450s 1A1 and 1A2 responsible for catalyzing retinol to retinal, the rate-limiting step for the biosynthesis of Retinoic acid.

$p < 0.05$

Modulation of retinoid binding & metabolizing genes is essential for Sytenol® A to function like Retinol

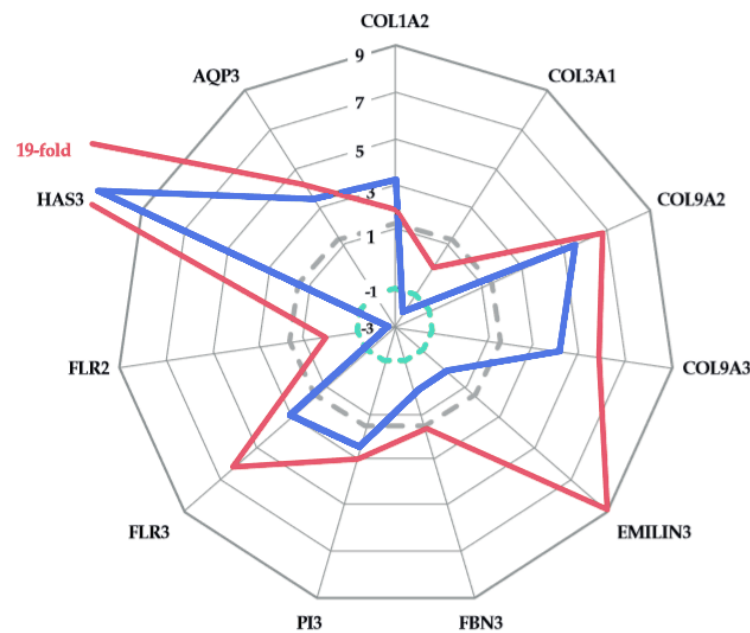
Sytenol[®] A: Modulator of Extracellular Matrix Gene

Gene expression profile: Fold-change vs Control

SEE PROTOCOL & GENE DETAILS

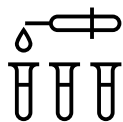
- Retinol
- Threshold of stimulatory activity
- Sytenol[®] A
- Threshold of stimulatory activity

$p < 0.05$



An intact and hydrated extracellular matrix genes is essential in maintaining youthful skin

Sytenol® A: Modulator of Extracellular Matrix Gene



Protocol

DNA microarray: Used full thickness Epiderm tissue from Mattek;
Considered expression profile of over ± 2 -fold change with $P < 0.05$

- > **Collagens:** Responsible for skin strength and elasticity; Degradation leads to wrinkles.
- > **Elastic Fibers:** (1) Elastin microfibril (EMILIN3) is a component of elastic fibers and localized mainly at the interface and is involved in the process of elastogenesis. (2) Fibrillins (FBN3): constitute the major backbone of multifunctional microfibrils in elastic and nonelastic extracellular matrices. Sytenol® A can maintain the desired level of Elastin required for maintaining the connective tissue homeostasis due to the up-regulation of elastase inhibitor (PI3).
- > **Fibronectin (FLR3; FLR2):** Maintains the shape of cells and is a prerequisite for matrix stability.
- > **Hyaluronan & Aquaporin:** Hyaluronans are responsible for the maintenance of a highly hydrated extracellular matrix in tissues & also involved in cell adhesion and supports cell migration. It is synthesized by one of three distinct hyaluronan synthases, such as HAS-3. Aquaporin 3 (AQP3) is the water/glycerol transporting channel protein expressed in the epidermis & helps maintain right levels of skin hydration, elasticity, and barrier recovery. AQP3 protein stimulation confirmed by using EpidermFT full thickness skin substitute.

$p < 0.05$

An intact and hydrated extracellular matrix genes is essential in maintaining youthful skin

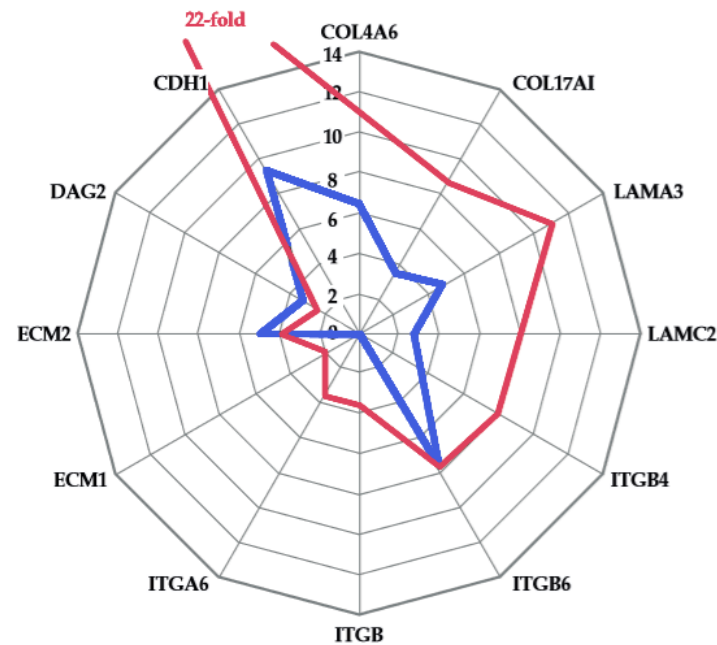
Sytenol® A: Modulator of Dermo-epidermal Junction Gene

Gene expression profile: Fold-change vs Control

SEE PROTOCOL & GENE DETAILS

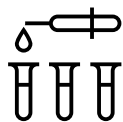
Retinol
Sytenol® A

$p < 0.05$



An intact basement membrane at the dermo-epidermal junction genes is essential in maintaining youthful skin

Sytenol® A: Modulator of Dermo-epidermal Junction Gene



Protocol

DNA microarray: Used full thickness Epiderm tissue from Mattek;
Considered expression profile of over ± 2 -fold change with $P < 0.05$

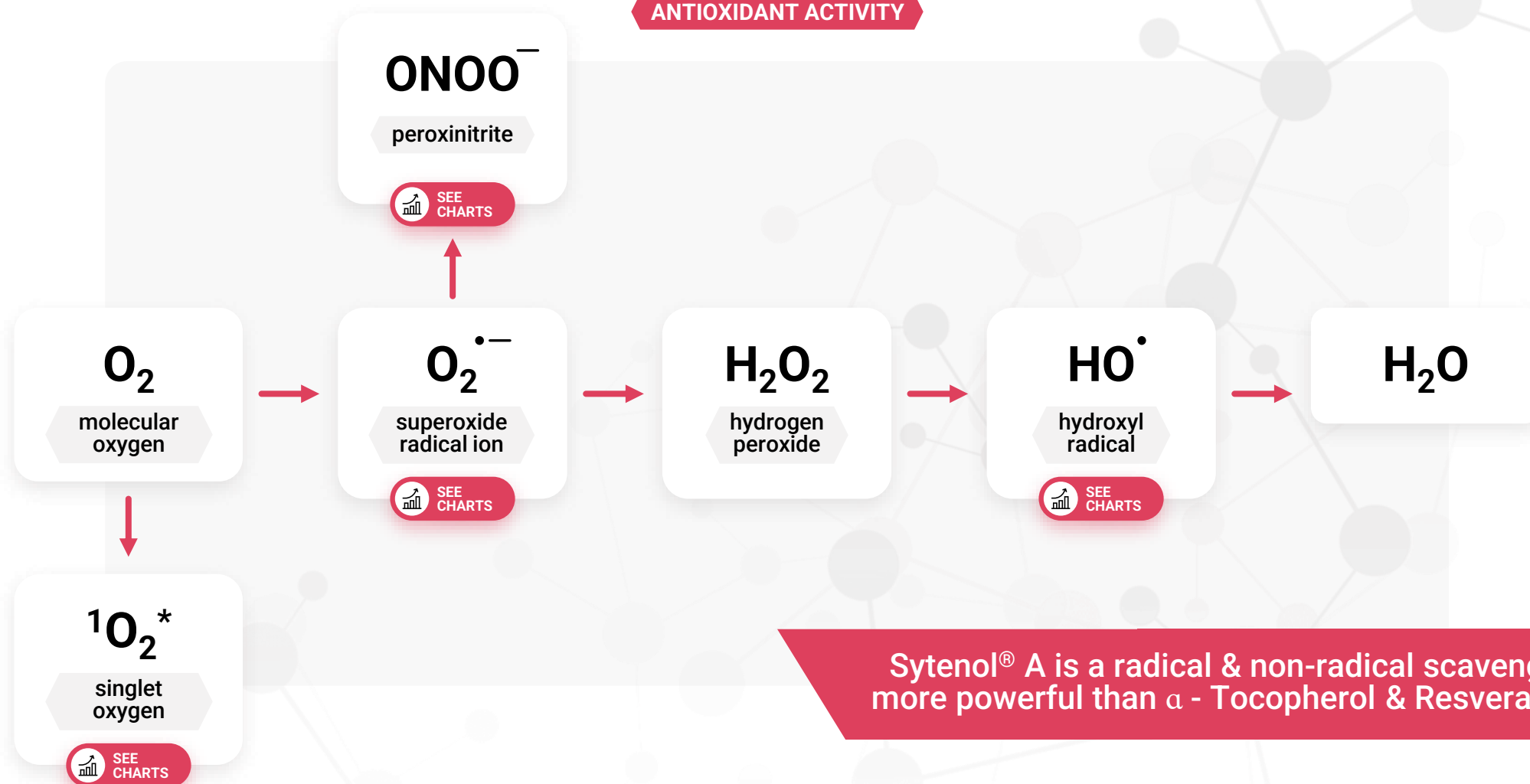
- > **Type IV Collagen:** Type IV Collagen forms supra molecular networks, which influence cell adhesion, migration, and differentiation.
- > **Type XVII Collagen:** Type XVII is one of the key anchoring fibrils that fortifies the attachment of the epidermis to the dermis.
- > **Laminin (LAMA3 and LAMC2):** Integral part of the structural scaffolding;; Major non-collagenous proteins in basal lamina. Involved in cell differentiation, migration, adhesion as well as phenotype and survival.
- > **E-Cadherin (CDH1):** Involved in tight junction formation & maintaining stem cell pluripotency; Sun over exposure causes loss of CDH1
- > **Integrin (ITGB4, ITGB6, ITGB8, ITGA6):** Defines cellular shape, mobility, and regulate the cell cycle; Key functions - attachment of the cell to the ECM & signal transduction from ECM to the cell.
- > **ECM Protein (ECM):** Involved in basement membrane and collagen fibril macro-assembly and growth factor binding.
- > **Dystroglycan (DAG2):** Involved in both assembly and maintenance of basement membrane structure; Loss causes disruption of epidermal differentiation.

$p < 0.05$

An intact basement membrane at the dermo-epidermal junction genes is essential in maintaining youthful skin

Sytenol® A: Broad Action Scavenger

ANTIOXIDANT ACTIVITY



antiaging menu >

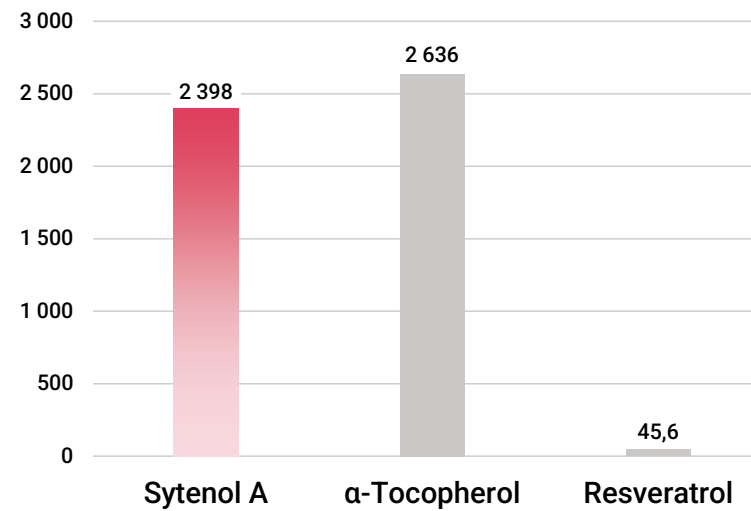
Sytenol® A is a radical & non-radical scavenger, more powerful than α - Tocopherol & Resveratrol

Sytenol® A: Broad Action Scavenger

ANTIOXIDANT ACTIVITY



$^1\text{O}_2^*$
Singlet oxygen
μmole Trolox
equivalent/g



O_2
molecular
oxygen

$^1\text{O}_2^*$
singlet
oxygen

SEE
CHARTS

H_2O

antiaging
menu

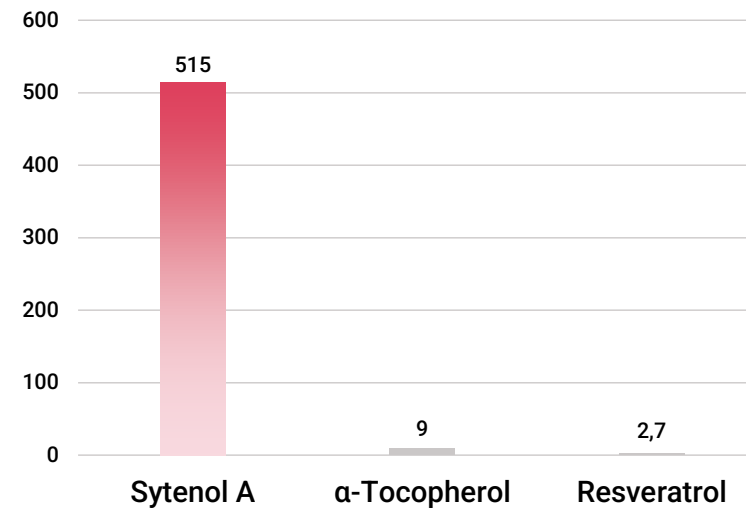
Sytenol® A is a radical & non-radical scavenger,
more powerful than α - Tocopherol & Resveratrol

Sytenol® A: Broad Action Scavenger

ANTIOXIDANT ACTIVITY

ONOO^-

ONOO^-
Peroxynitrite
μmole Trolox
equivalent/g



O_2
molecular
oxygen

$^1\text{O}_2^*$
singlet
oxygen

SEE
CHARTS

H_2O

antiaging
menu

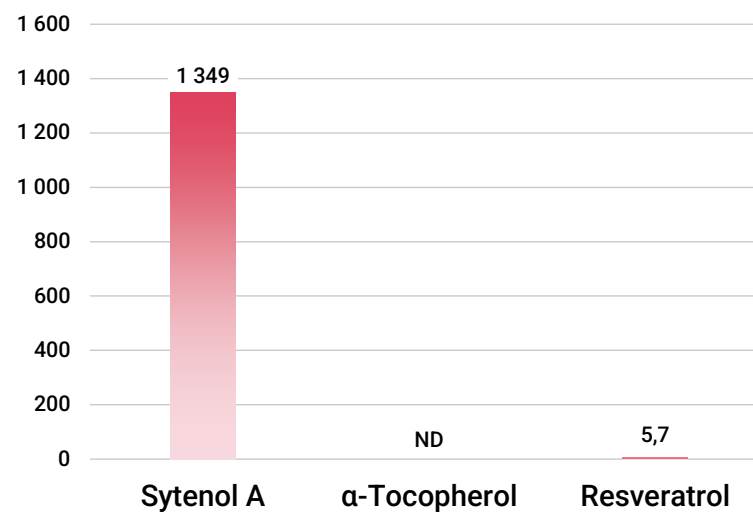
Sytenol® A is a radical & non-radical scavenger,
more powerful than α - Tocopherol & Resveratrol

Sytenol® A: Broad Action Scavenger

ANTIOXIDANT ACTIVITY



O₂^{•-}
Superoxide radical ion
 μmole Trolox equivalent/g



O₂
 molecular oxygen

¹O₂^{*}
 singlet oxygen

SEE CHARTS

H₂O

antiaging menu

Sytenol® A is a radical & non-radical scavenger, more powerful than α - Tocopherol & Resveratrol

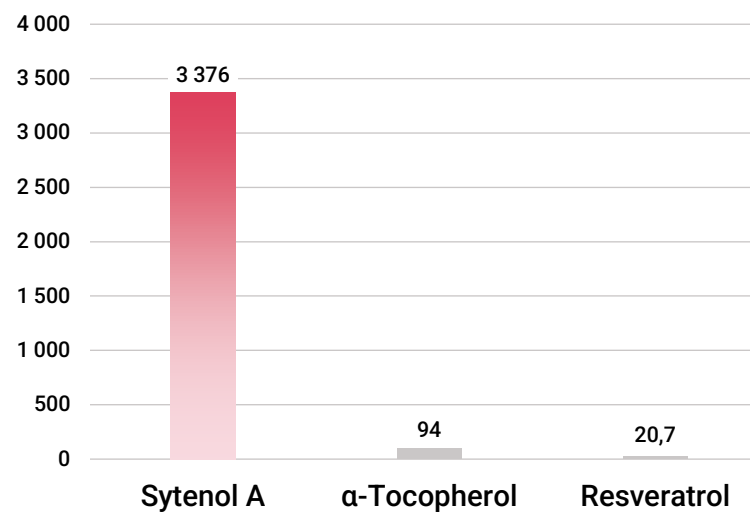
Sytenol® A: Broad Action Scavenger

ANTIOXIDANT ACTIVITY



Hydroxyl radical

μmole Trolox equivalent/g



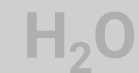
molecular oxygen



singlet oxygen



SEE CHARTS



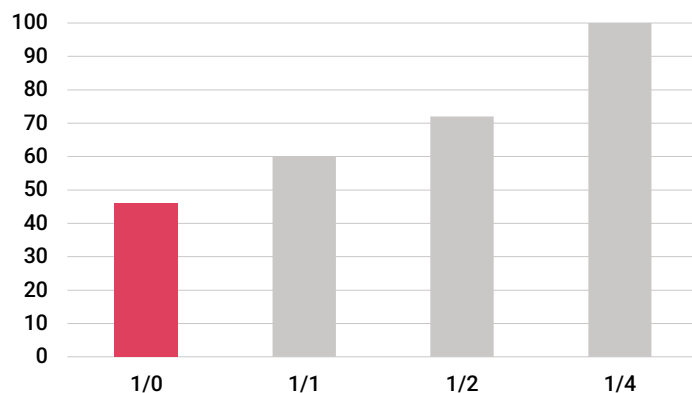
antiaging menu

Sytenol® A is a radical & non-radical scavenger, more powerful than α - Tocopherol & Resveratrol

Sytenol® A: Stabilizer for Retinol

Under Photo-oxidative Environment

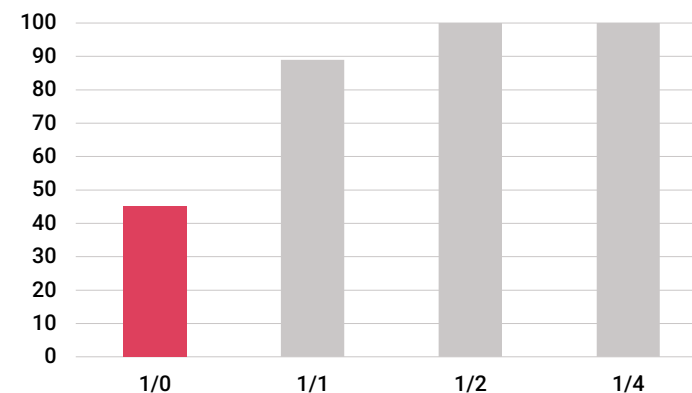
% Improvement in Retinol stabilization under photo-oxidative environment with different ratios of Sytenol® A



SEE PROTOCOL

Under Singlet Oxygen Environment

% Improvement in Retinol stabilization with different ratios of Sytenol® A

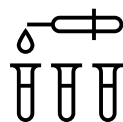
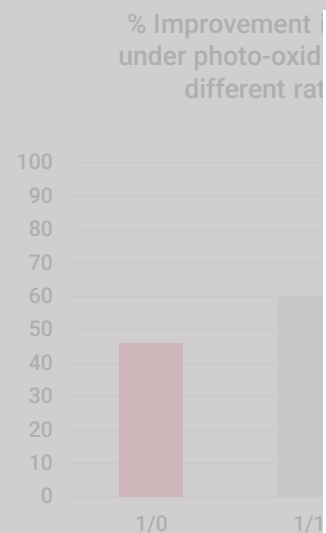


SEE PROTOCOL

There is a complete photostabilization of Retinol using four-fold excess of Sytenol® A, and a protection under singlet oxygen environment with a two-fold excess of Sytenol® A

Sytenol® A: Stabilizer for Retinol

Under Photo-oxidative Environment



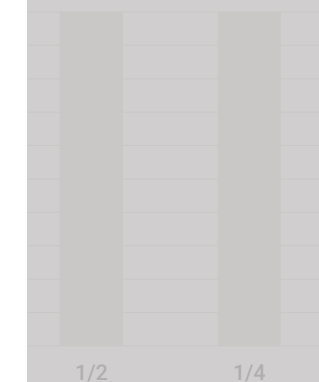
Protocol

- > Retinol (50 µg/ml) was dissolved in ethanol and then Sytenol® A was added 50, 100 & 200 µg/ml using 5mL OPTICLEAR® vials (KIMBLE glass, Chicago Heights, IL)
- > Test vials were then placed in a Rayonet RPR-100 photochemical reactor equipped with four RMR-3000 (UVB) and four RMR 3500 (UVA) lamps (Southern New England Ultraviolet Company, Branford, CT), the combination of UVA and UVB simulated day light condition
- > The samples were irradiated at 31 °C for 5 min at a dose of 13 J/cm².
- > Retinol was quantified by HPLC using a 4.6 × 250 mm Luna C 18 column (Phenomenex, Torrance, CA) equipped with a DAD detector at 280 nm
- > The mobile phase consisted of 75% A (acetonitrile) and 25% B (methanol), isocratically ran at 1 ml/min for 10 minutes.

Under Singlet Oxygen Environment



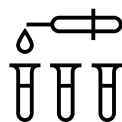
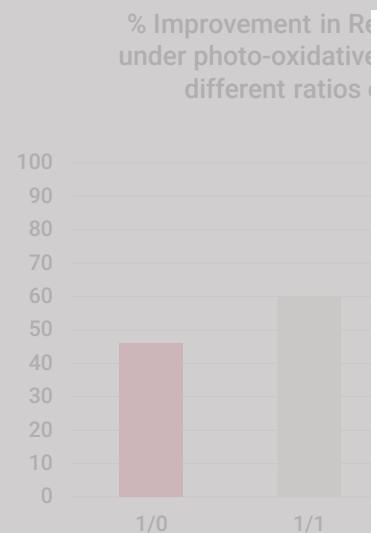
stabilization
of Sytenol® A



There is a complete photostabilization of Retinol using four-fold excess of Sytenol® A, and a protection under singlet oxygen environment with a two-fold excess of Sytenol® A

Sytenol® A: Stabilizer for Retinol

Under Photo-oxidative Environment



Protocol

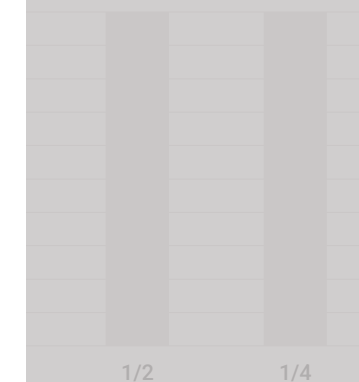
Reference - *Tetrahedron Letters* 43 (2002) 8731–8734

- > Retinol (50 µg/ml) was dissolved in ethanol and then Sytenol® A was added 50, 100 & 200 µg/ml using 5mL OPTICLEAR® vials.
- > Singlet oxygen was generated by addition of H₂O₂ and Lithium molybdate in pH 9.01.
- > The mixture was then incubated at 37°C in dark for 15 hrs. During the period, singlet oxygen was generated and retinol was oxidized.
- > Retinol was quantified by HPLC on a 4.6 × 250 mm Luna C 18 column (Phenomenex, Torrance, CA) using a DAD detector with detection 280 nm.
- > The mobile phase consisted of 75% A (acetonitrile) and 25% B (methanol), isocratically ran at 1 ml/min.

Under Singlet Oxygen Environment



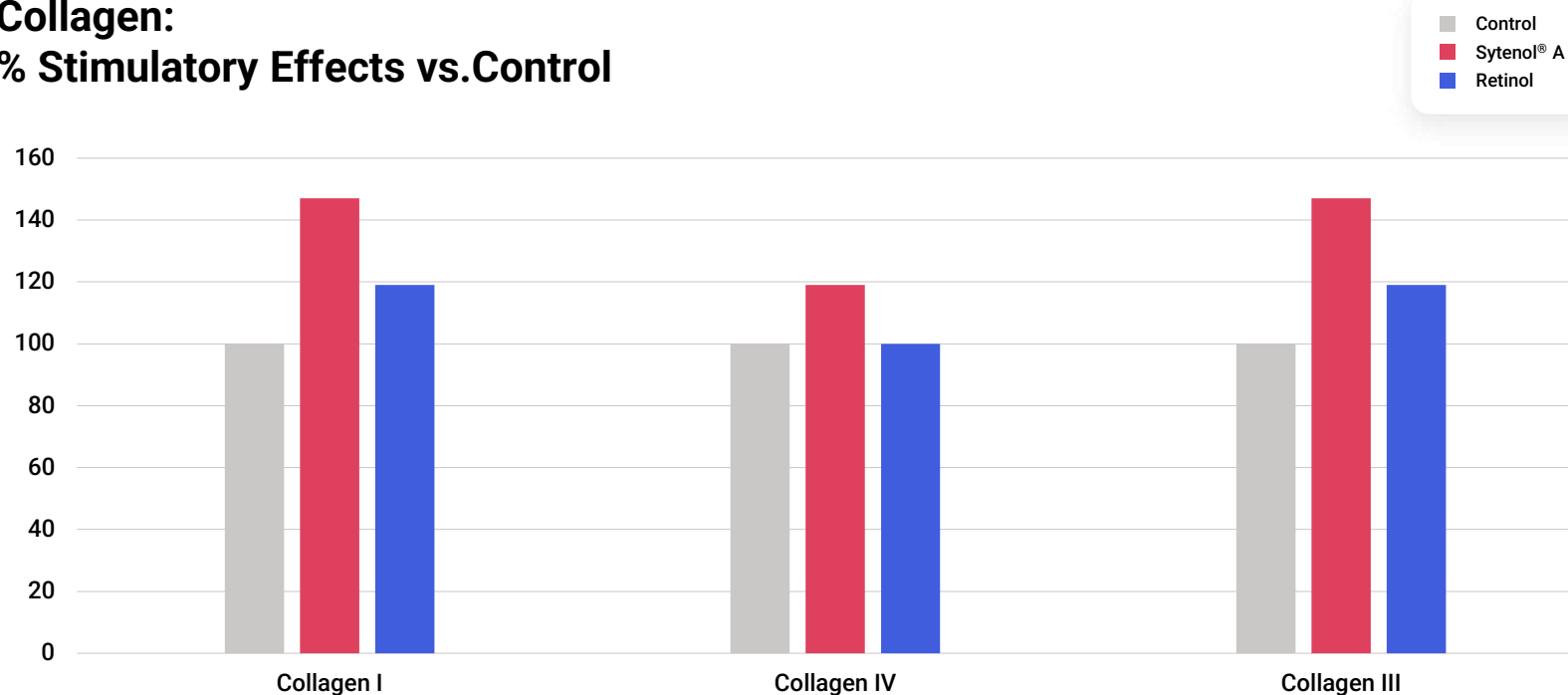
Retinol stabilization
ratios of Sytenol® A



There is a complete photostabilization of Retinol using four-fold excess of Sytenol® A, and a protection under singlet oxygen environment with a two-fold excess of Sytenol® A

Sytenol® A: Collagen Stimulator

Collagen:
% Stimulatory Effects vs. Control



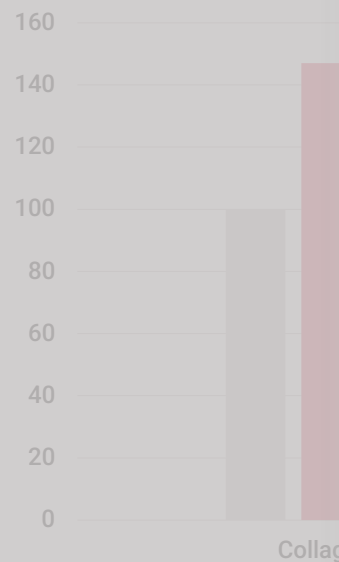
✓ SEE PROTOCOL

in vivo >

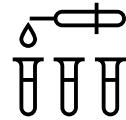
Sytenol® A significantly stimulates collagen from neonatal (type I & IV collagens) or mature (type III collagen) fibroblasts

Sytenol® A: Collagen Stimulator

Collagen: % Stimulatory Effect

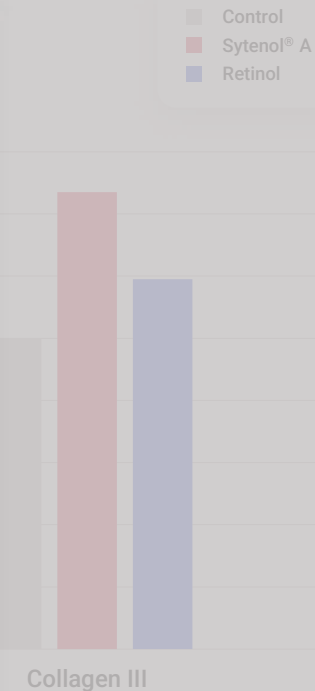


SEE PROTOCOL



Protocol

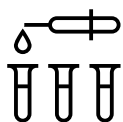
- > **Cell Lines:** Normal human dermal fibroblasts (COL I & IV)
Fibroblasts from 68 yrs old (COL III)
No cell proliferation seen with Sytenol® A & Retinol at 10 µg/ml
- > **Cell Culture:** DMEM with 5% calf serum
- > **Use level:** 10 µg/ml; dissolved in DMSO & then diluted in water; Water used as a control; Cells treated with the products for 7 days
- > **Protocol:** Tested for type I, III & IV collagen by sandwich ELISA using affinity-purified antibodies, followed by streptavidin-avidin-HRP conjugate and ABTS (ELISA)
- > **Quantification:** Used BioRad microplate spectrophotometer 3550-UV at 405nm with background subtraction at 660nm and analyzed with Microplate Manager v.2 software for Macintosh (BioRad)



in vivo >

Sytenol® A significantly stimulates collagen from neonatal (type I & IV collagens) or mature (type III collagen) fibroblasts

Double-blind Assessment of Topical Sytenol® A and Retinol for Facial Photoaging



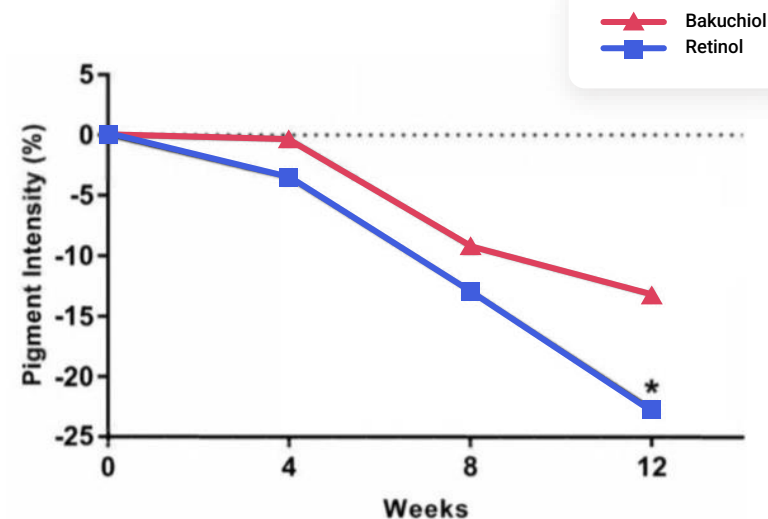
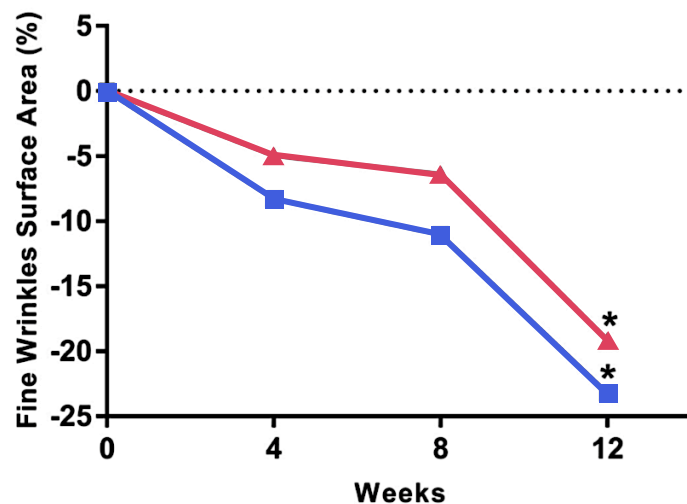
Protocol

> Human volunteers: 44 subjects Age – 47±7.2 years	> Study duration: 12 weeks; March to November 2017	> Test substances: Cream with 0.5% Sytenol®A twice daily or 0.5% retinol daily	> Test sites: Full face
> Results: Compared to initial day (pre-treatment)	> Statistical Analysis: Statistical significance defined as p=0.05 or less	> Methodology: A facial photograph and analysis system was used to obtain and analyze high-resolution photographs of subjects at 0, 4, 8, and 12 weeks. Subjects also completed tolerability assessment questions to review side effects. During study visits, a board-certified dermatologist, blinded to study group assignments, graded pigmentation and redness.	

Work done by University of California - Davis



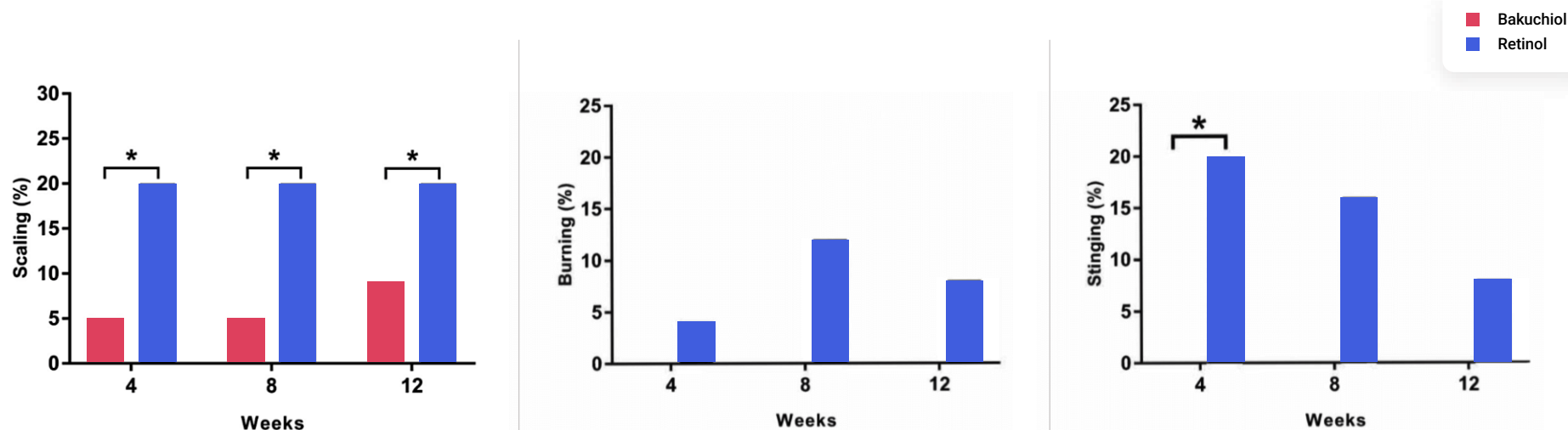
Results: Comparison between Topical Sytenol® A and Retinol for Facial Photoaging



Sytenol® A and retinol both significantly decreased wrinkle surface area and hyperpigmentation, with no statistical difference between the compounds

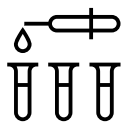
$p \leq 0.05$

Results: Comparison of Side Effects between Topical Sytenol® A and Retinol



There were more subjective reports of itching and burning in the retinol group, though this was not statistically significant. There were more statistically significant subjective reports of scaling in the retinol group

Second Anti-aging Clinical Study



Protocol

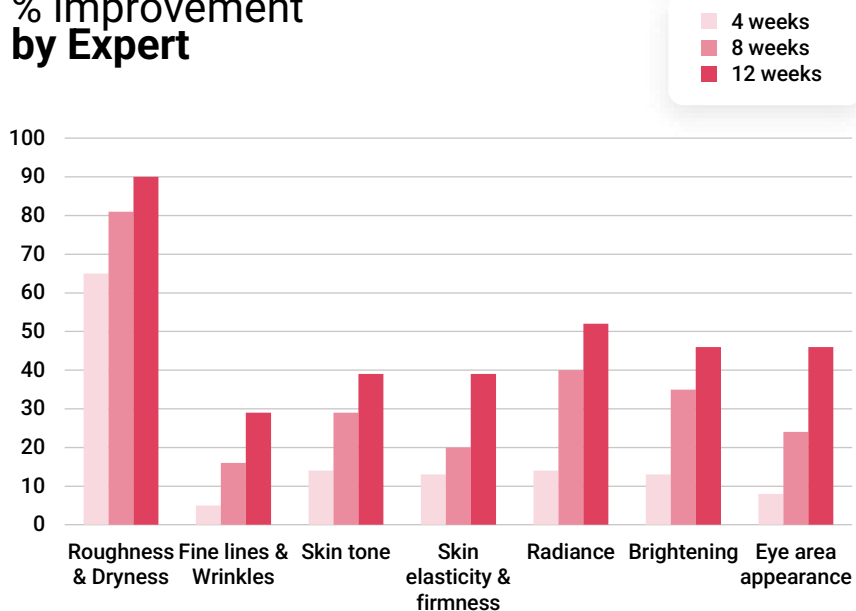
> Human volunteers: 17; 16 Completed; Caucasian (14), Hispanics (2); Age: 41 to 60 yrs	> Study duration: 12 weeks; May to August 2012 (winter in Australia)	> Test substances: Lotion with 0.5% Sytenol A; Contains No sunscreen and No moisturizer	> Test sites: Full face
> Results: Compared to initial day (pre-treatment)	> Statistical Analysis: Statistical significance defined as p=0.05 or less	> Methodology: Expert grading/Self-assessment by panelists (Grading 0 to 4): (1) Roughness & Dryness; (2) Fine lines & wrinkles; (3) Skin tone; (4) Skin elasticity & firmness; (5) Radiance; (6) Brightening; (7) Overall eye-area appearance; Silicone Replica Analysis: (1) Wrinkle depth & (2) Skin roughness; Photography: Before & after the treatments; Readings were taken at baseline, 4, 8 & 12 weeks.	
	> Application frequency : About 2 g twice a day		

Work done by Cantor Research Lab, New York

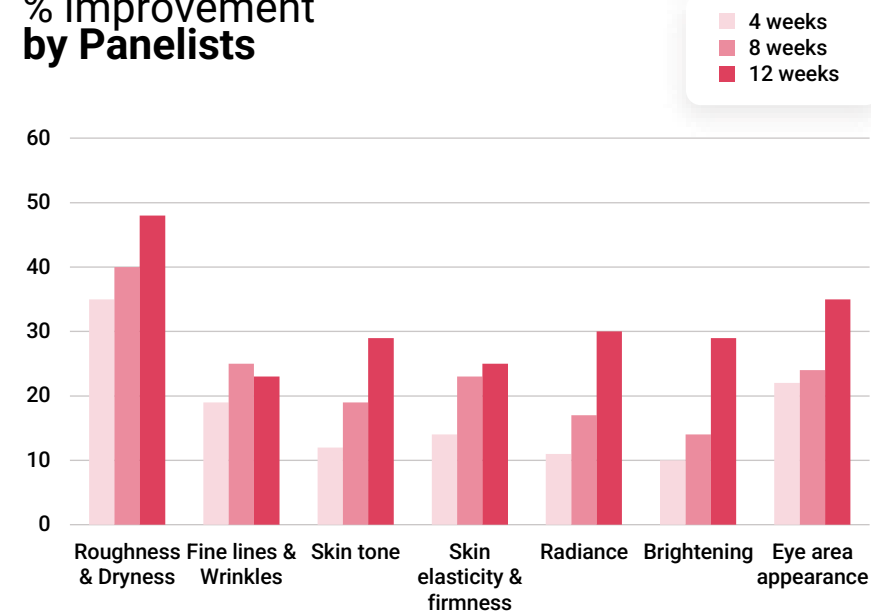


Anti-aging Study Results: % Improvement Seen by Experts & Panelists

% improvement
by Expert



% improvement
by Panelists



$p \leq 0.05$

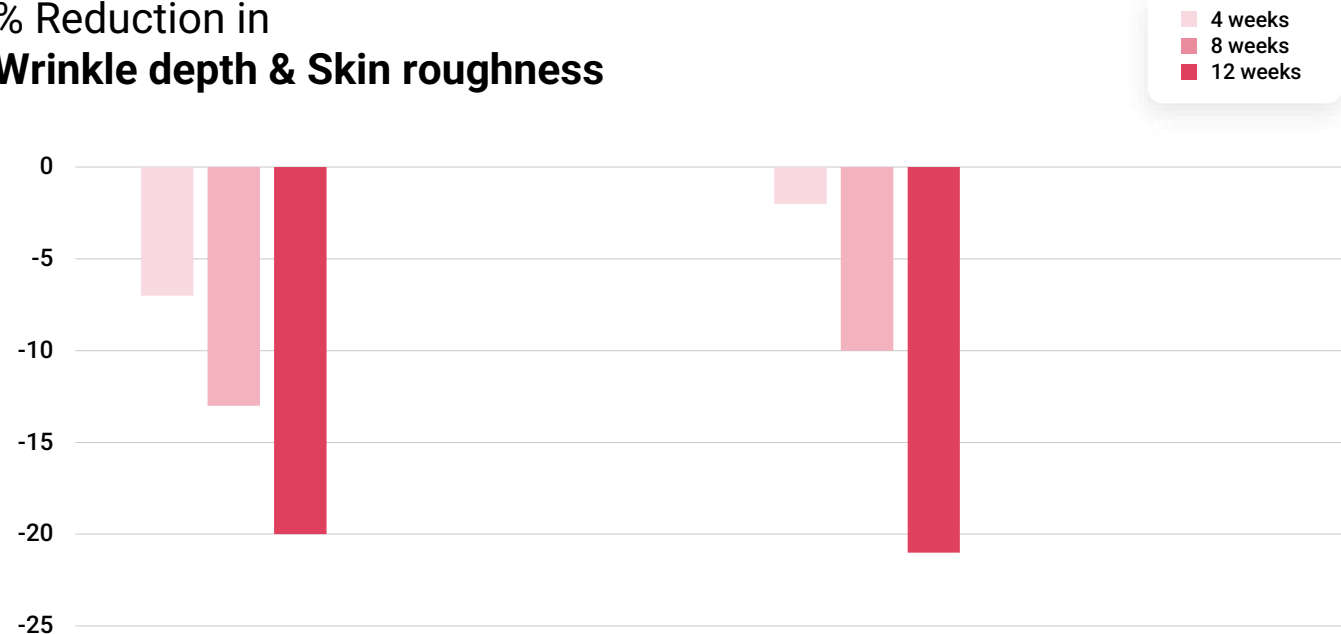
After twelve weeks treatment, significant improvement in lines and wrinkles, pigmentation, elasticity, firmness and overall photo-damage was observed both by experts and panelists, with no irritating effect on skin

$p \leq 0.05$



Anti-aging Study Results: Silicone Replica Analysis

% Reduction in Wrinkle depth & Skin roughness



✓ SEE PROTOCOL

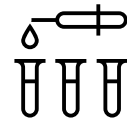
After twelve weeks treatment, significant improvement in wrinkle depth and skin roughness is confirmed by silicone replica analysis



Anti-aging Study Results: Silicone Replica Analysis

% Reduction in
Wrinkle d

0
-5
-10
-15
-20
-25



Protocol

- > At each visit, a single silicone replica was made of the target area and a photographic record was kept of this target for subsequent relocation.
- > The samples were stored in controlled conditions for comparative measurement. Comparative analysis of skin profilometry was conducted using surface roughness and wrinkle depth analysis.
- > The heights of the replicated wrinkles were measured using Miyomoto Surf test profilometer. Ry (depth) and Ra (mean roughness) were recorded at each time of measuring operation. The area scanned from each sample was clearly mapped so as to determine the same area in respective of Week 4, Week 8, and Week 12 samples.

$P \leq 0.05$



4 weeks
8 weeks
12 weeks



SEE PROTOCOL

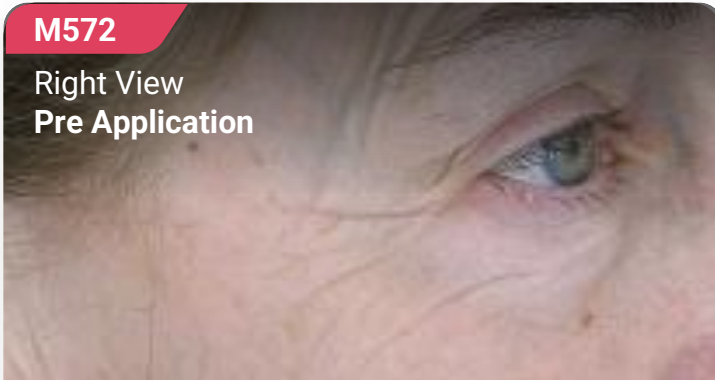
After twelve weeks treatment, significant improvement in wrinkle depth and skin roughness is confirmed by silicone replica analysis



Anti-aging Study Results: Photographic Comparison Before & After Treatment

M572

Right View
Pre Application



Wrinkle Depth (μm)

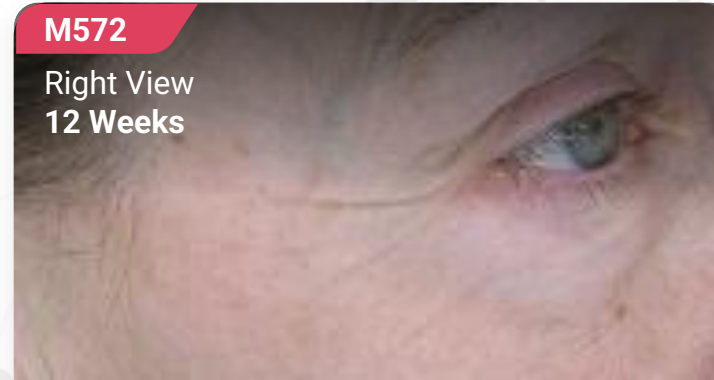
BASE	4 WK	8 W	12 W
18.50	18.30	13.80	12.70

Skin Roughness (μm)

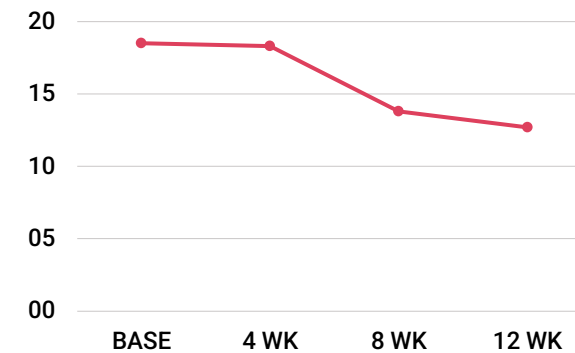
BASE	4 WK	8 W	12 W
3.10	4.05	2.91	2.42

M572

Right View
12 Weeks



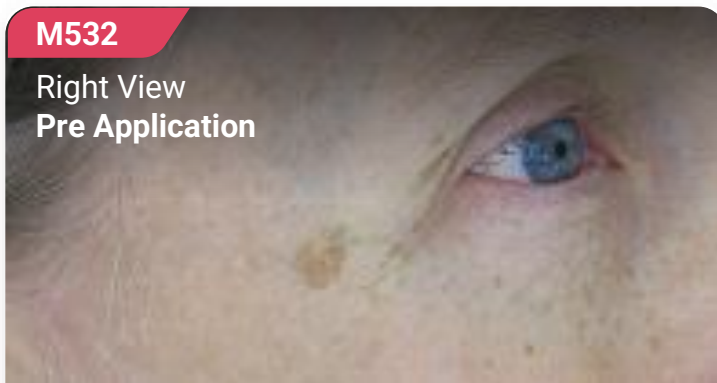
Change in Wrinkle depth (μm)



Anti-aging Study Results: Photographic Comparison Before & After Treatment

M532

Right View
Pre Application



Wrinkle Depth (μm)

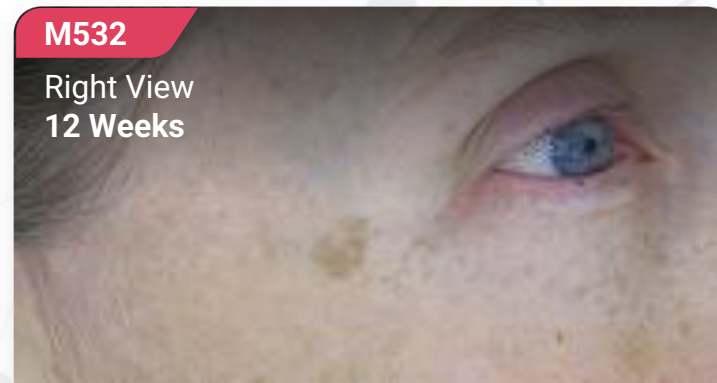
BASE	4 WK	8 W	12 W
38.60	30.40	29.80	23.90

Skin Roughness (μm)

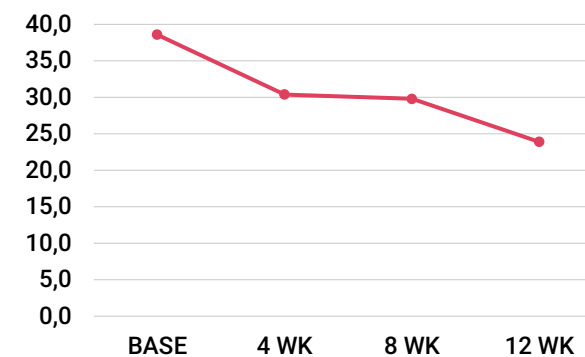
BASE	4 WK	8 W	12 W
9.15	8.35	8.61	5.07

M532

Right View
12 Weeks



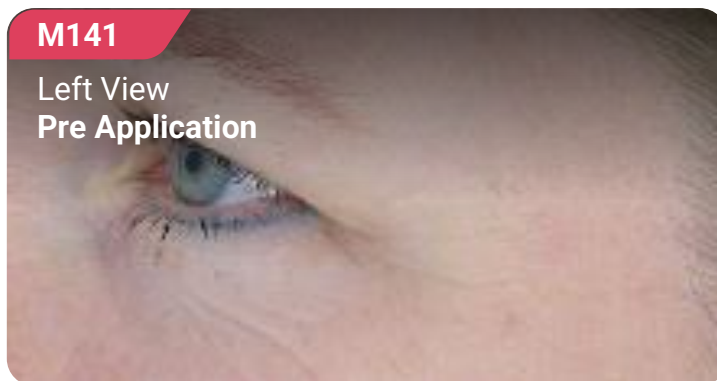
Change in Wrinkle depth (μm)



Anti-aging Study Results: Photographic Comparison Before & After Treatment

M141

Left View
Pre Application



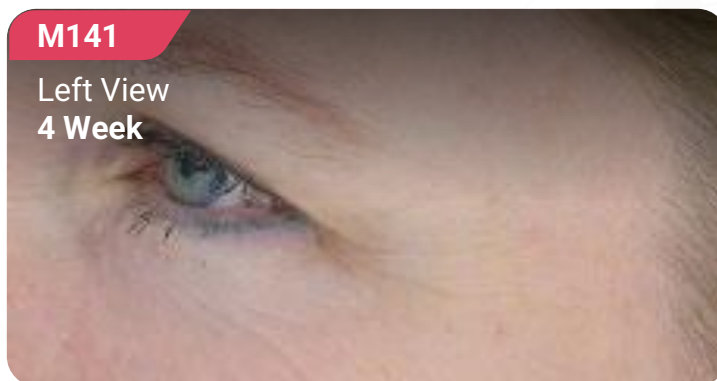
M141

Left View
8 Week



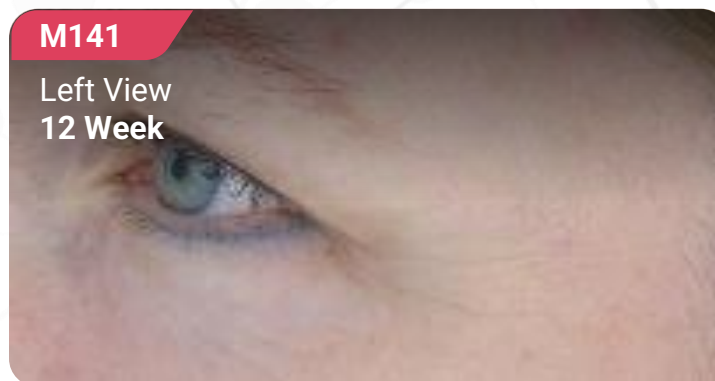
M141

Left View
4 Week



M141

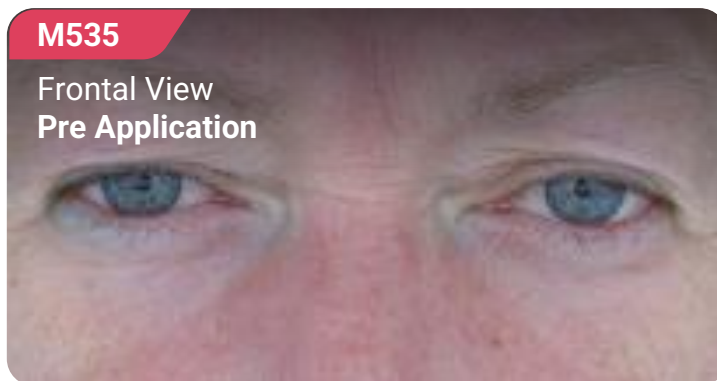
Left View
12 Week



Anti-aging Study Results: Photographic Comparison Before & After Treatment

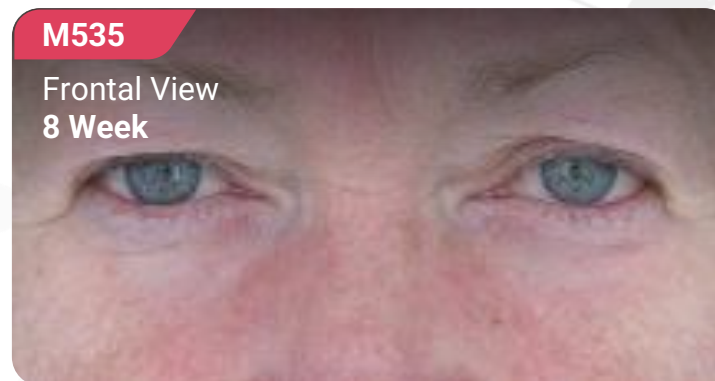
M535

Frontal View
Pre Application



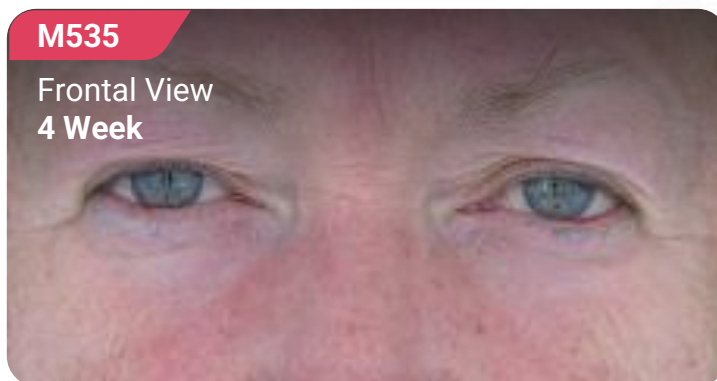
M535

Frontal View
8 Week



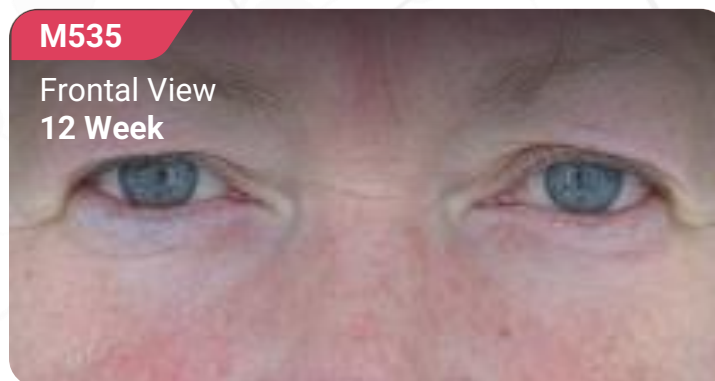
M535

Frontal View
4 Week



M535

Frontal View
12 Week



Sytenol® A is Effective Against 4 Factors Involved in Acne

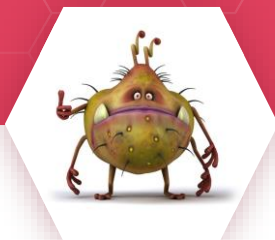
1



Sebum Production
& its integrity



2



Overpopulation
of Bacteria



3



Inflammation



4



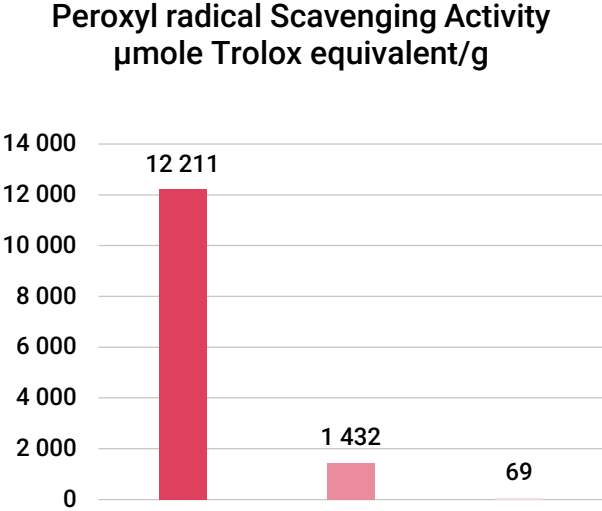
Hyperkeratinization



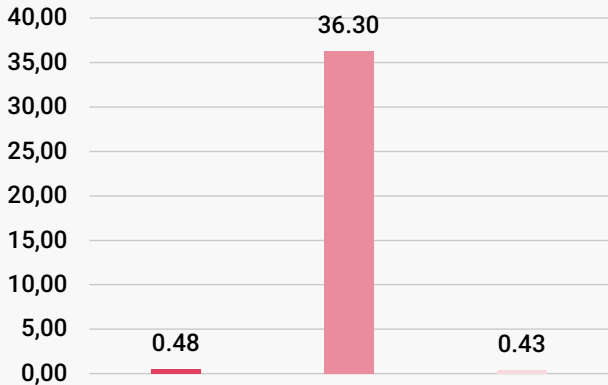
Sytenol® A Helps to Maintain the Integrity of Skin Lipids

Squalene is a polyunsaturated lipid and highly susceptible to peroxidation and photodegradation.

Squalene peroxides promote acne, roughening of skin, wrinkling & pigmentation.



Lipid peroxidation inhibitory activity (IC50)
in μg/ml



■ Sytenol® A
■ α-Tocopherol
■ Resveratrol

SEE PROTOCOL

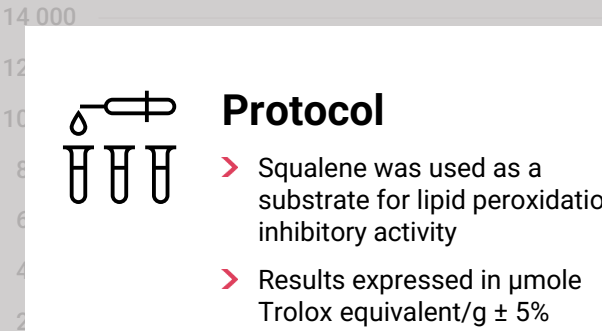
Sytenol® A is a radical & non-radical quencher. It is 60-fold more effective than Tocopherol to protect skin against lipid peroxidation.

Sytenol® A Helps to Maintain the Integrity of Skin Lipids

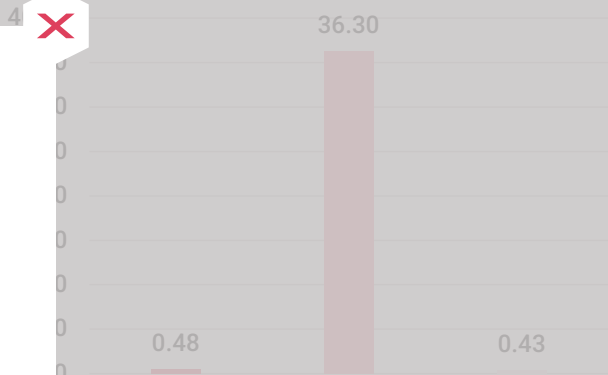
Squalene is a polyunsaturated lipid and highly susceptible to peroxidation and photodegradation.

Squalene peroxides promote acne, roughening of skin, wrinkling & pigmentation.

Peroxyl radical Scavenging Activity
μmole Trolox equivalent/g



Lipid peroxidation inhibitory activity (IC50)
in μg/ml



Protocol

- > Squalene was used as a substrate for lipid peroxidation inhibitory activity
- > Results expressed in μmole Trolox equivalent/g ± 5%

Sytenol® A
α-Tocopherol
Resveratrol

SEE PROTOCOL

Sytenol® A is a radical & non-radical quencher. It is 60-fold more effective than Tocopherol to protect skin against lipid peroxidation.

Sytenol® A Helps to Control Bacterial Overpopulation

- > Overwhelming body of evidence implicates propionobacteria, and P. acnes in particular, in inflammatory acne. P. acnes is neither a primary pathogen nor a bystander.
- > P acnes plays an active role in determining the nature and extent of the immune response.

MIC values in µg/ml

Micro organisms	Sytenol® A	Retinol	Benzoyl Peroxide	Salicylic acid
P. acnes	1.5	31	50	>100
Staphylococcus aureus	1	N/D	15.6	N/D
Staphylococcus epidermidis	2	N/D	>100	N/D
Candida albicans	1.5	N/D	N/D	N/D

Treatments that reduce P. acnes numbers lead to clinical improvement of acne.

Acne-affected skin has also been reported to have higher levels of Staphylococcus and Candida.

Sytenol® A has broad-action antibacterial and antifungal property

Sytenol® A: Inflammation Control

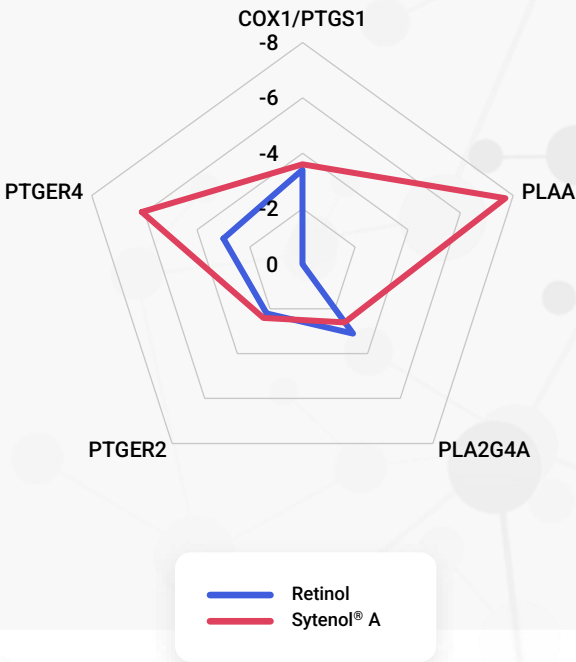
Sytenol® A is effective in inhibiting

- > COX-1 (IC50 14.7 µg/ml) and COX-2 (IC50 514 µg/ml)
- > 5-lipoxygenase (IC50 23.5 µm)

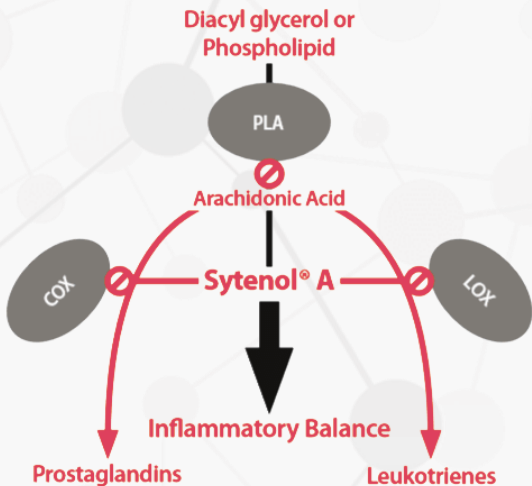
P ≤ 0.05

SEE PROTOCOL

Comparison of inhibition of inflammatory genes



Inhibition of pro-inflammatory enzymes



Sytenol® A: Inflammation Control

Comparison of inhibition of inflammatory genes

Inhibition of pro-inflammatory enzymes

Sytenol® A
is effective in inhibiting

- > COX-1 (IC50 14.7 µg/ml)
and COX-2 (IC50 514 µg/ml)
- > 5-lipoxygenase
(IC50 23.5 µm)

$P \leq 0.05$



SEE PROTOCOL

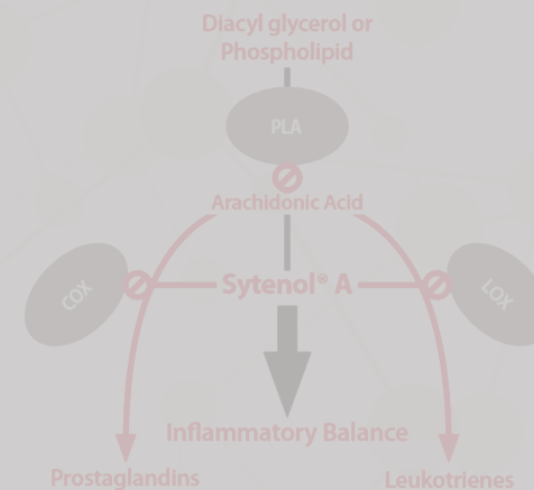
$P \leq 0.05$



Protocol

**DNA microarray test using full
thickness Epiderm tissue (Mattek)**

- > Retinol and Sytenol® A were dissolved in DMSO, further diluted in water and applied to full thickness Epiderm tissues
- > Incubation for two days
- > Selected gene expressions having $\geq \pm 2$ -fold change
- > Statistical significance: $p = \leq 0.05$

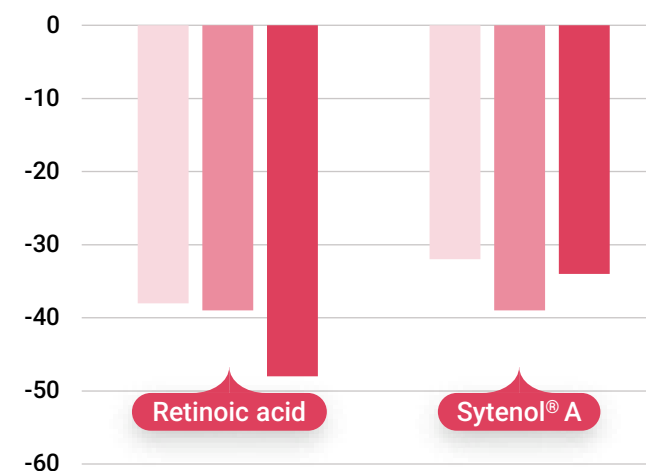


Sytenol® A Helps to Correct Hyperkeratinization

- > One of the most crucial initial events in the development of acne lesions is hyperkeratinization in the follicular infundibulum and sebaceous duct resulting in microcomedones.
- > Follicular keratinization seems to be triggered by relative deficiency of linoleic acid and generation of peroxides in sebum.
- > Increased 5- α -reductase (DHT) may act on infundibular keratinocytes leading also to abnormal hyperkeratinization

% Down-regulates 5- α -reductase expression

SEE PROTOCOL



5 µg/ml 10 µg/ml 20 µg/ml

in vivo >

At 10 µg/ml, Sytenol® A down regulates 5- α -reductase by about 40% and compares well with retinoic acid

Sytenol® A Helps to Correct Hyperkeratinization

- > One of the most crucial initial events in the development of acne lesions is hyperkeratinization of the follicular infundibulum. This leads to a plugged sebaceous duct resulting in microcomedones.
- > Follicular keratinization can be triggered by relative deficiency of linoleic acid and generation of peroxides in sebum.
- > Increased 5- α -reductase activity may act on infundibular keratinocytes leading to abnormal hyperkeratinization.

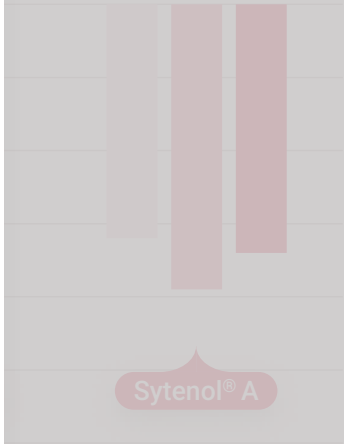


Protocol

- > **Cell:**
HaCat human keratinocytes
- > **Incubation time:**
48 hrs
- > **Quantification:**
Immunofluorescence technique; Determined spectrophotometrically at 485/518 nm Ex/Em

% Down-regulates 5- α -reductase expression

SEE PROTOCOL



in vivo >

At 10 µg/ml, Sytenol® A down regulates 5- α -reductase by about 40% and compares well with retinoic acid

Anti-Acne Clinical Study #1



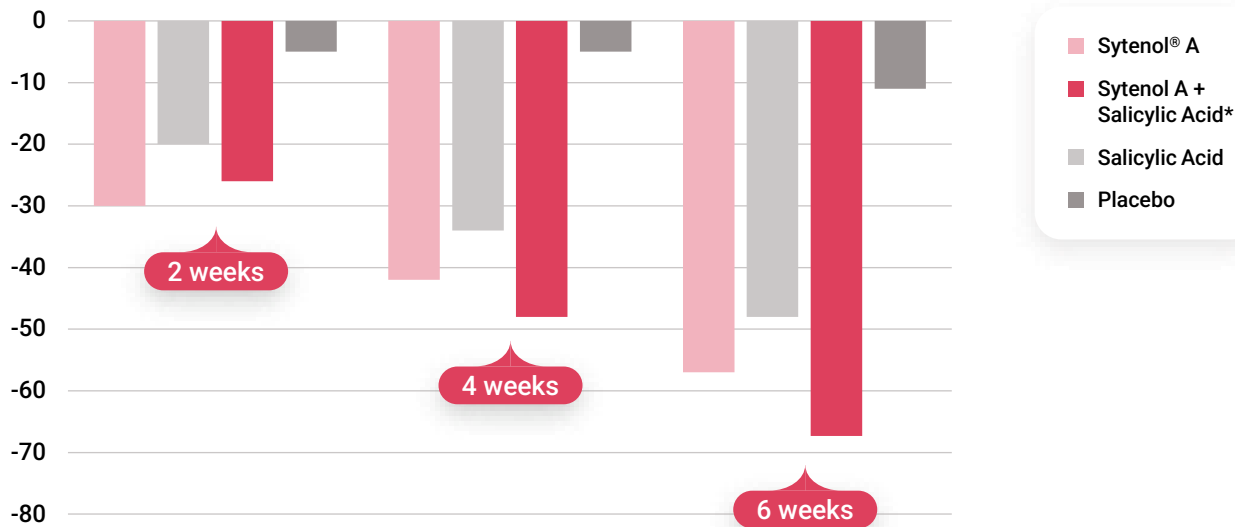
Protocol

> Subjects: 15 x 4 with mild (up to 10 comedones), moderate (10 to 25) to severe (>25) acne	> Age: 16 to 30	> Duration of study: 6 weeks	> Frequency of application: Full face
> Assessment of Efficacy: • Acne score at initial day • Reduction in comedones after 2, 4 & 6 weeks of treatment • Reduction in erythema and pruritus after 2, 4 & 6-weeks of treatment	> Products: • Sytenol® A (1% lotion), • Salicylic acid (2% lotion), • Sytenol® A + Salicylic acid (1+2% lotion) • Placebo (no active)		



Results Anti-Acne Clinical Study #1

% Reduction in Acne:
Global Acne Grading



p ≤ 0.05

*** A SPF is recommended**
when combining Sytenol® A
+ salicylic acid due to the
sun sensitization with
salicylic acid

Sytenol® A alone or in combination with Salicylic acid provide statistically significant improvement in acne-affected skin and is well tolerated

Anti-Acne Clinical Study #2



Protocol

> Subjects: 10 with mild (up to 10 comedones), moderate (10 to 25) acne	> Age: 14 to 55	> Duration of study: 6 weeks	> Frequency of application: Twice a day
> Assessment of Efficacy: • Acne score at initial day • Reduction in comedones after 2, 4 & 6 weeks of treatment • Reduction in erythema and pruritus after 2, 4 & 6-weeks of treatment • Photography initial and after 6 weeks of treatment • $p \leq 0.05$			> Products: • Lotion with 1% Sytenol® A

$p \leq 0.05$

Anti-Acne Clinical Study #2 Panelist ID 7, Left



Anti-Acne Clinical Study #2 Panelist ID 9, Left





Sytenol® A Potential Applications for Skincare



Natural
Natrue
Vegan
ISO16128 : 1

Retinol-like
Activity

Broad action
Antioxidant

Protection against
Lipid Peroxidation

Multifunctional
Anti-aging

Applications

> Anti-aging

- Reduce fine lines and wrinkles
- Improve overall skin texture & tone
- Suitable for sensitive skin

> Anti-acne

- Reduce comedones and spots in only 2 weeks
- Improve the quality of the sebum



Formulation Tips

Suggested use level

- > 0.5 to 1.0% w/w

Formulation choices

- > Skin care, antiaging lotion, anti-acne products

Advices

- > Add Sytenol® A to the formulation after making emulsion with a processing temperature of about 50°C.
- > Alternately, Sytenol® A can be added directly to the oil phase. Avoid prolonged heating above 75°C.
- > Miscible in a wide variety of emollients, solubilizers, such as, Caprylic/capric triglycerides, C 12-15 alkyl benzoates, ethyl linoleate, mineral oils, jojoba oils, olive oils, ...
- > Contact with iron or copper compounds must be avoided as it is a phenolic compound, which by nature tend to become colored in the presence of metal ions; Addition of a small amount of chelating agent (~0.05%) resolves this problem.
- > Airless/opaque packaging is recommended to protect product integrity and stability over time
- > Finished product must be acidic: pH<6.5 is recommended

Formulation examples

[SEE DETAIL](#)

Age Defying Day Cream
with Sytenol® A and HydraSynol® DOI

[SEE DETAIL](#)

Anti-Aging Night Cream
with Sytenol® A and HydraSynol® DOI

[SEE DETAIL](#)

Acne-Control Lotion
with Sytenol® A and Salicylic Acid





Age Defying Day Cream with Sytenol® A and HydraSynol® DOI

(Formulation# RC-01-42.02)

INCI NAME	TRADE NAME/SUPPLIER	% W/W
Phase A		
Water		74.35
Disodium EDTA	RonaCare Disodium EDTA/EMD	0.10
Glycerin	Glycerin/Rita	2.00
Butylene Glycol	Jeechem BUGL/Jeen	2.00
D Panthenol	Ritapan DL, 50%/Rita	0.50
Water & Potassium Sorbate & Sodium Benzoate	Euxyl K 712/Schulke	1.00
Sodium Hydroxide & Water	Sodium Hydroxide 50% Aq. Sln./Sigma Aldrich	0.20
Phase B		
Xanthan Gum	Keltrol® CG-T/CP Kelco	0.10
Acrylates/C10-30 Alkyl Acrylate Crosspolymer	Carbopol Ultrez 21/Lubrizol	0.30
Phase C		
Simmondsia Chinensis (Jojoba Oil)	Biochemica Jojoba Oil Deodorized/Hallstar	3.00
Butyrospermum Parkii (Shea Butter)	Shebu Refined/Rita	1.50
Behenyl Alcohol	Lanette 22/BASF	1.25
Potassium Cetyl Phosphate	Amphisol® K/DSM	1.00
Stearic Acid	Stearic Acid/Protameen	1.00
Dimethicone	Dimethicone Fluid/Making Cosmetics	5.00
Beeswax	Natural Beeswax/Rita	1.50
Isosorbide Dicaprylate	HydraSynol® DOI/Sytheon	2.00
Tocopheryl Acetate	Vitamin E Acetate/Jeen	0.20
Ammonium Acryloyldimethyltaurate/VP Copolymer	Aristoflex AVC/Clariant	0.30
Glyceryl Stearate	Protachem GMS-D/Protameen	1.50
Bakuchiol	Sytenol® A/Sytheon	1.00
Phase D		
Lavandula Angustifolia (Lavender) Oil	Lavender Oil/Jeen	0.20
Total		100.00



Procedure

Weigh Phase A in the main kettle equipped with a homogenizer. Sprinkle in Phase B. Once dispersed, heat to 70- 75°C. Weigh Phase C in a side kettle equipped with a propeller mixer. Heat to 70-75°C. Once both phases are at the proper temperature, add Phase C to Phase AB. Mix for 15-20 minutes. Switch to side sweep mixing and start cooling. Add Phase D at 50°C. Finish cooling the batch to room temperature.



Notes

pH value: 6.0-6.5

Viscosity: Spindle-TF, S96; Speed-0.3 rpm (Brookfield RVT, Spindle C, 10rpm) at 25°C

Range: 2,500,000-2,800,000 mPas



BACK FORMULATION TIPS



Anti-Aging Night Cream with Sytenol® A and HydraSynol® DOI

(Formulation#RC01-13)

INCI NAME	TRADE NAME/SUPPLIER	% W/W
Phase A1		
Water	Water (demineralized)	QS
Disodium EDTA	Versene Na/Dow	0.10
Glycerin	Emery 916/BASF	3.00
Butylene Glycol	Butylene Glycol/Ruger	2.00
D Panthenol	Panthenol/BASF	0.50
Phase A2		
Xanthan Gum	Keltrol® CG-T/CP Kelco	0.15
Carbomer	Carbopol Ultrez 21/Lubrizol	0.40
Sucrose Stearate	Systema SP50C/MMP	2.50
Poly C8-22 Alkyl Acrylate/Metacrylic Acid Cross Polymer	Intelimer 8600/Air Products	2.50
Phase B		
Simmondsia Chinensis (Jojoba Oil)	Jojoba Oil/Jeen Intl.	3.00
Shea Butter	Shebu Refined/Rita	2.50
Behenyl Alcohol	Lanette 22/BASF	1.50
Potassium Cetyl Phosphate	Amphisol® K/DSM	1.00
Stearic Acid	Pristerene® 4911/Croda	1.00
Dimethicone	Dow Corning 200,100 cst/Dow Corning	4.00
Natural Beeswax	Beeswax/Rita	2.00
Isosorbide Dicaprylate	HydraSynol® DOI/Sytheon	3.00
Alpha Tocopherol	Vitamin E/BASF	0.20
Hydrogenated Polyisobutene	Panalane L14E/Lipo	4.00
Bakuchiol	Sytenol® A/Sytheon	0.50
Phase C		
Sodium Hydroxide	Sodium Hydroxide (50%sol)	0.30
Phase D		
Fragrance	Women's FragranceK3258/Carruba	0.20
Red 33 (CI 17200)	D&C Red 33 (1% sol)	0.05
Phenoxyethanol, Ethylhexyglycerine	Euxyl PE 9010/Schulke	1.00
Total		100.00



Procedure

Combine A1; disperse A2 one by one in A1 while stirring and heat A to 75°C. Combine ingredients of phase B and heat to 75°C. Add phase B to A with good mixing. Homogenize mixture at moderate speed for 3-5 min while adding phase C (adjust pH to 5.5-6.0). Cool batch to 40°C with propeller agitation until mixture is homogeneous. Add phase D ingredients while continue mixing.



Notes

pH value: 5.5-6.0

Viscosity: 80,000-100,000mPas
(Brookfield RVT, Spindle C, 10rpm) at 25°C

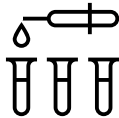


BACK FORMULATION TIPS

Acne-Control Lotion with Sytenol® A and Salicylic Acid

(Formulation#RC-01-11.02)

INCI NAME	TRADE NAME/SUPPLIER	% W/W
Phase A		
Water		72.20
Disodium EDTA	RonaCare Disodium EDTA/EMD	0.10
Glycerin	Glycerin/Rita	3.00
Butylene Glycol	Jeechem BUGL/Jeen	2.00
Caprylyl Glycol & Phenoxyethanol & Hexylene Glycol	Jeecide Cap-2/Jeen	1.00
Polysorbate 20	Ritabate 20/Rita	1.00
Sodium Hydroxide & Water	NaOH, 50% Aq. Sln./Sigma Aldrich	0.80
Phase B		
Ethoxydiglycol	Transcutol CG/Gattefosse	4.50
Salicylic Acid	Salicylic Acid/Novacyl	2.00
Phase C		
Xanthan Gum	Vanzan NF/Vanderbilt	0.50
Phase D		
Dicaprylyl Ether	Cetiol OE/BASF	4.00
Cetyl Alcohol	Cetyl Alcohol/Protameen	1.50
Tocopheryl Acetate	Vitamin E Acetate/DSM	0.20
Glyceryl Stearate, PEG-100 Stearate	Protachem GMS-165/Protameen	3.00
Cetearyl Alcohol, Ceteareth-20	Ritapro 300/Rita	1.00
Beeswax	Natural Beeswax/Rita Corp	1.00
Bakuchiol	Sytenol® A/Sytheon	1.00
Phase E		
Polyacrylamide, C13-14 Isoparaffin, Laureth-4	Sepigel 305/Seppic	2.00
Total		100.00



Procedure

Weigh Phase A in the main kettle equipped with a homogenizer. Weigh Phase B in a side kettle and mix until all solids are dissolved. Sprinkle Phase C into Phase A. Once dispersed, add pre-mixed Phase B and heat to 70-75°C. Weigh Phase D in a side kettle equipped with a propeller mixer. Heat to 70-75°C. Once both phases are at the proper temperature, add Phase D to Phase ABC. Mix for 15-20 minutes. Add Phase E and mix for 5 minutes. Switch to side sweep mixing and cool the batch to room temperature.



Notes

pH value: 4.0-4.5
Viscosity: Spindle-TE, S95 Speed-0.3 rpm
Range: 500,000-900,000 mPas



Publications

Bakuchiol may be another natural solution to reverse the course of nature

ML Levy - *British J Dermatology*, 180:253-254, 2019

Prospective, randomized, double-blind assessment of topical bakuchiol and retinol for facial photoaging

S Dhaliwal et al. - *British J Dermatology*, 180:289-296, 2019

Bakuchiol: A retinol-like functional compound modulating multiple retinol and non-retinol targets

RK Chaudhuri - *Cosmeceuticals and Active Cosmetics, 3r Edition, Eds., RK Sivamani, J Jagdeo, P Elsner & HI Maibach, Francis & Taylor, Boca Raton, Chapter 1, 1-18, 2015*

Bakuchiol to stabilize retinol and polyunsaturated lipids,

RK Chaudhuri & B Ou - *Cosm & Toil*, 130:64-75, 2015

Bakuchiol: A retinol-like functional compound revealed by gene expression profiling and clinically proven to have anti-aging effects

RK Chaudhuri & K Bojanowski - *International J Cosmetic Science*, 36(3):221-230, 2014

Bakuchiol in the management of acne-affected skin

RK Chaudhuri & F Marchio - *Cosm & Toil*, 126:502-510, 2011

The Miracle of Retinol: Why is there still so much fuss about retinoids and retinoid-like products?

RK Chaudhuri - *SPC*, 23-24, May 2010



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